

ALC-Experiment Nr. B31

Superconductivity

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Group Nr. M602

Pravin Mani Kannan Barathraj Radhakrishnan Subramanian
Mat. Nr: 23459332 Mat. Nr: 23403835



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1 Introduction

In 1902, Lord Kelvin raised a question that would shape low-temperature physics for the next decade: what happens to the electrical resistance of a metal as temperature approaches absolute zero? Would the reduced thermal motion of lattice ions eventually allow electrons to pass without scattering, driving resistance to zero, or would the electrons themselves freeze, sending resistance toward infinity?^[1] Neither outcome could be tested without access to temperatures far below anything achievable at the time.

This changed on 10 July 1908, when Kamerlingh Onnes successfully liquefied helium, opening up temperatures below 14 K for the first time^[1]. After early attempts with platinum were hampered by instrumentation limitations, **mercury** was selected as the test material due to the high purity achievable by distillation. It was later found that purity played no significant role in the onset of superconductivity, as the resistance vanished identically in pure mercury and impure amalgams alike, with lead and tin also found to be superconductors shortly after, at transition temperatures near 6.2 K and 4.2 K respectively. On 8 April 1911, cooling mercury below 4.2 K caused its resistance to drop so sharply it fell below the detection limit of the apparatus entirely^[1:2].

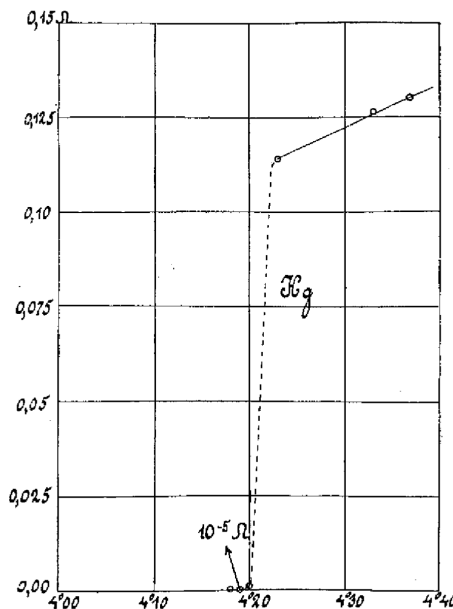


Figure 1: Ultra-cooled mercury showing nearly zero resistance^[1].

To quantify the effect, Onnes induced persistent currents in closed **superconducting rings** and monitored them over time. Any finite resistance R would cause exponential decay as

$$I(t) = I_0 e^{-\frac{R}{L}t} \quad (1.1)$$

In 1933, Meissner and Ochsenfeld further established that superconductors expel magnetic flux entirely from their interior upon cooling below T_c , identifying ideal diamagnetism as a second and independent defining property of the superconducting state^[2].

These two phenomena form the experimental basis of this report, in which the superconducting transition of $\text{YBa}_2\text{Cu}_3\text{O}_7$ and $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+x}$ samples is characterised through **resistive four-point measurements** and inductive AC-susceptibility measurements, alongside a demonstration of the Meissner-Ochsenfeld effect through magnetic levitation^[3].

2 Theoretical Framework

Superconductivity is characterised by the complete vanishing of electrical resistance below a material-specific critical temperature, T_c , at which point a persistent current can flow indefinitely without dissipation. Unlike an ordinary metallic conductor, whose resistance decreases only gradually as temperature is lowered, a superconductor undergoes an **abrupt drop to zero resistance at T_c** (see Figure 1). The mechanisms governing electrical transport differ fundamentally between metals, semiconductors, and superconductors, and each class exhibits a qualitatively distinct $R(T)$ behaviour. The microscopic basis for superconducting behaviour is provided by BCS theory (Bardeen, Cooper, and Schrieffer, 1957), which describes the formation of Cooper pairs through phonon-mediated electron - electron attractions, producing a dissipation-free quantum superfluid that carries current without resistance^[4].

3 Electrical Resistance and Temperature Dependence

3.1 Metals

In a metal, electrical conduction is carried by a gas of free electrons occupying states up to the Fermi energy E_F . In the **Drude model**, a constant electric field accelerates electrons between collisions, and the resulting steady-state current density gives the conductivity^[5]

$$\sigma = \frac{ne^2\tau}{m_e} \quad (3.1)$$

where n is the free electron density, e is the electron charge, m_e is the electron mass, and τ is the mean time between scattering events. The electrical resistivity follows as

$$\rho = 1/\sigma = m_e/ne^2\tau \quad (3.2)$$

Two scattering mechanisms contribute independently to the total resistivity^[5]. The first arises from thermally excited lattice vibrations (phonons), which become stronger with increasing temperature. The second comes from static crystal imperfections such as impurities and vacancies, which are temperature-independent. These two contributions add independently according to **Matthiessen's rule**

$$\rho(T) = \rho_L(T) + \rho_i \quad (3.3)$$

where ρ_i is the residual resistivity at $T \rightarrow 0$, reflecting the purity and structural quality of the sample^[5]. At high temperatures $\rho_L \propto T$, where ρ_L is the phonon contribution. So resistance increases linearly. As $T \rightarrow 0$, phonon scattering freezes out.

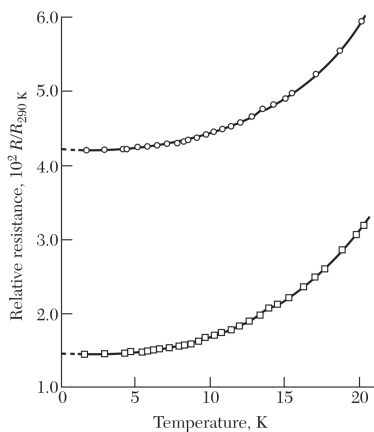


Figure 2: Relative resistance of potassium for two impurity concentrations having different residual resistivities ρ_i at $T = 0$. Adapted from Kittel^[5].

3.2 Semiconductors

In a semiconductor, the valence and conduction bands are separated by an energy gap E_g of order 1 eV^[5]. At zero temperature no free carriers exist and the material is effectively insulating. As temperature increases, electrons are **thermally excited across the gap**, generating mobile electrons in the conduction band and holes in the valence band, both of which contribute to conduction. The intrinsic carrier concentration grows exponentially with temperature as

$$n_i \propto T^{3/2} \exp\left(\frac{-E_g}{2k_B T}\right) \quad (3.4)$$

and since this exponential factor dominates, the resistivity decreases with increasing temperature^[5]

$$\rho(T) \propto \exp\left(\frac{E_g}{2k_B T}\right) \quad (3.5)$$

This is the opposite behaviour to metals. Since $\sigma = 1/\rho$, one equivalently plots $\ln \sigma$ against $1/T$ (refer Fig. 3b), which yields a straight line with slope $-E_g/2k_B$, providing a direct way to extract the band gap from conductance measurements^[5].

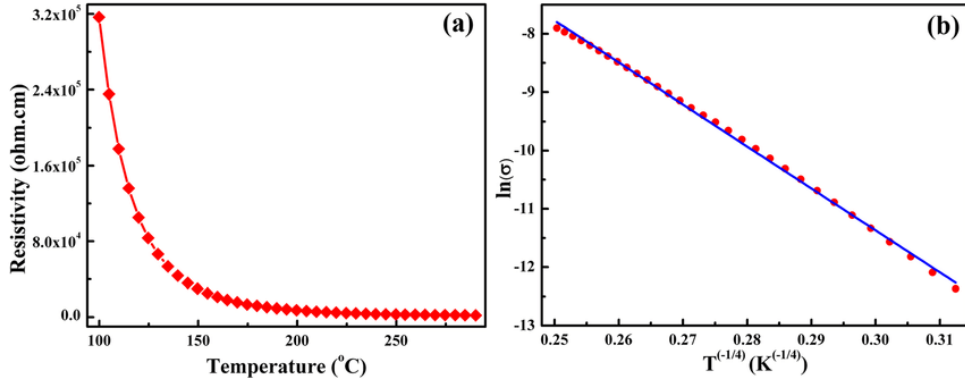


Figure 3: (a) Resistivity vs temperature for silicon, polyaniline (PANI), and a polyaniline-silicon nanocomposite (PNC). (b) $\ln \sigma$ vs $1/T^{1/4}$ showing the linear behaviour characteristic of semiconductors. Adapted from Applied Physics Letters (2013)^[6].

3.3 Superconductors and BCS Theory

Above T_c a superconductor behaves as a normal metal. Below T_c , however, the resistance drops abruptly to zero, which cannot be explained by reduced phonon scattering alone. The **Bardeen-Cooper-Schrieffer (BCS)** theory provides the microscopic explanation: a phonon-mediated attractive interaction between electrons near the Fermi surface causes them to form bound pairs known as Cooper pairs^[7]. Each Cooper pair consists of two electrons with equal and opposite momenta and opposite spins. For instance, if one-particle orbital is occupied by a spin up electron $\uparrow$ and wavevector $\mathbf{k}$ then the other is occupied by a spin down electron $\downarrow$ with wavevector $-\mathbf{k}$. If the orbital is vacant then its vice versa.^[5]

All Cooper pairs condense into the same macroscopic quantum ground state, described by a single wave function with a well-defined phase^[2]. This condensate is separated from excited states by an energy gap Δ . A thermal fluctuation cannot scatter the condensate **unless it supplies enough energy to break a Cooper pair**, which requires $k_B T \sim \Delta$. Below T_c this condition is not met, so scattering is suppressed entirely and resistance vanishes^[7].

$$2\Delta(0) = 3.52 k_B T_c \quad (3.6)$$

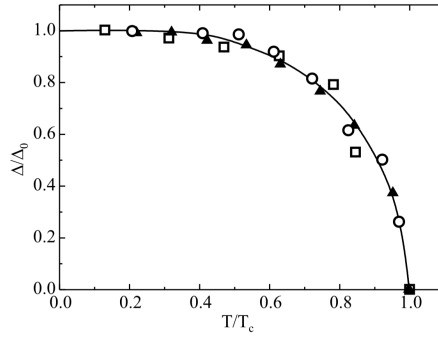


Figure 4: Temperature dependence of the Δ [8].

The BCS theory predicts a universal relation between the zero-temperature gap and the critical temperature. Experimental values across many materials typically fall in the range $3.0 k_B T_c$ to $4.5 k_B T_c$, in good agreement with the BCS prediction [7].

4 Magnetic Properties of Superconductors

4.1 Ideal Diamagnetism and the Meissner Effect

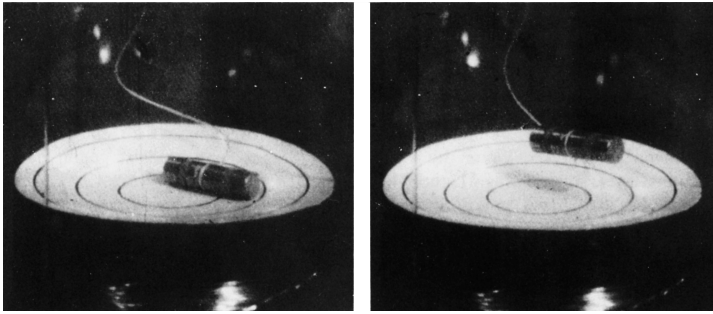


Figure 5: Demonstration of the Meissner-Ochsenfeld effect by dropping a permanent magnet onto a lead bowl. Left: $T > T_c$, right: $T < T_c$. Adapted from Buckel and Kleiner [2].

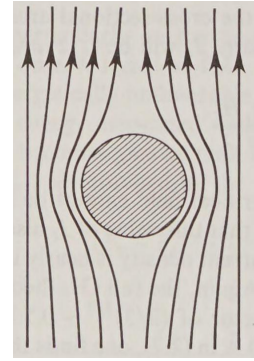


Figure 6: Magnetic field lines around a superconductor showing diamagnetic behavior [7].

A superconductor cooled below its critical temperature T_c exhibits perfect diamagnetism, spontaneously expelling all internal magnetic flux in what is known as the **Meissner-Ochsenfeld effect**, first observed by Meissner and Ochsenfeld in 1933 [2]. The external field does not vanish abruptly at the surface but instead penetrates a thin surface layer over a characteristic length scale known as the London penetration depth, defined as

$$\lambda_L = \sqrt{\frac{m}{\mu_0 q^2 n_s}}, \quad (4.1)$$

where m is the carrier mass, q the charge, n_s the superfluid density, and μ_0 the permeability of free space. Beyond this surface layer the field decays exponentially as

$$B(x) = B(0) \exp\left(-\frac{x}{\lambda_L}\right), \quad (4.2)$$

becoming negligibly small deep within the bulk. It is worth noting that this behaviour is fundamentally different from that of a perfect classical conductor, and can only be understood within a quantum mechanical framework [2;7].

4.2 Type-I vs. Type-II Superconductors

Type-I superconductors are characterised by a **single critical field** H_c , below which ideal diamagnetism is maintained and above which superconductivity is abruptly destroyed; elemental superconductors such as mercury and lead fall into this category^[2;7]. Type-II superconductors instead exhibit **two critical fields**, H_{c1} and H_{c2} , and are distinguished by a Ginzburg-Landau parameter

$$\kappa = \frac{\lambda_L}{\xi_{GL}} > \frac{1}{\sqrt{2}}, \quad (4.3)$$

where ξ_{GL} is the Ginzburg-Landau coherence length, which gives rise to a negative surface energy between normal and superconducting regions^[9;7]. Between H_{c1} and H_{c2} , the material enters the Shubnikov phase, in which **magnetic flux partially penetrates the sample as discrete flux lines** while zero electrical resistance is maintained^[2]. Notably, virtually all superconducting alloys and compounds discovered since the 1960s, including all known high-temperature superconductors where H_{c2} can exceed 100 T, belong to the Type-II class^[9;2].

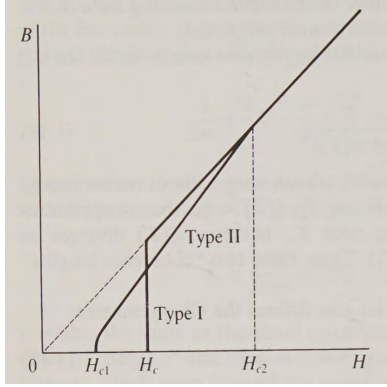


Figure 7: Comparison of flux penetration behavior of type I and type II superconductors with the same thermodynamic critical field H_c , where $H_{c2} = \sqrt{2} \kappa H_c$ ^[7]

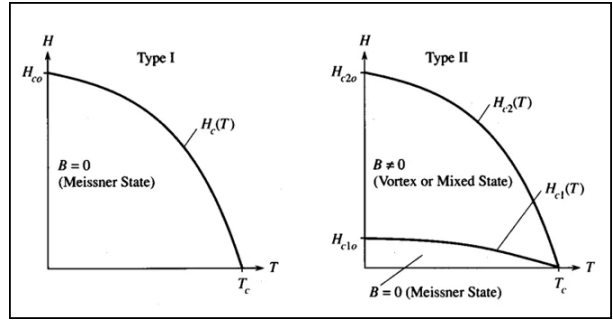


Figure 8: Magnetic phase diagram of Type-II superconductor shows the existence of **vortex states or Shubnikov phase**^[10].

4.3 Flux Quantization and the Abrikosov Vortex Lattice

In the Shubnikov phase, magnetic flux does not penetrate uniformly but is instead confined to discrete quantised tubes, each carrying exactly one magnetic flux quantum

$$\Phi_0 = \frac{h}{2e} \approx 2.07 \times 10^{-15} \text{ T m}^2, \quad (4.4)$$

as confirmed experimentally by Doll and Nabauer, and independently by Deaver and Fairbank, in 1961^[2]. Each vortex consists of a normal-state core of radius ξ_{GL} , which physically represents the spatial extent of a Cooper pair, surrounded by circulating supercurrents that decay over the length scale λ_L ^[7]. The lower and upper critical fields can be estimated as

$$H_{c1} \approx \frac{\Phi_0}{\pi \lambda_L^2}, \quad H_{c2} \approx \frac{\Phi_0}{\pi \xi_{GL}^2}, \quad (4.5)$$

showing that a **smaller coherence length gives rise to a substantially larger upper critical field**. Using Ginzburg-Landau theory, Abrikosov showed that these vortices arrange into a periodic triangular lattice that minimises the free energy of the mixed state, a prediction later confirmed through neutron diffraction and iron-colloid decoration experiments^[9;2].

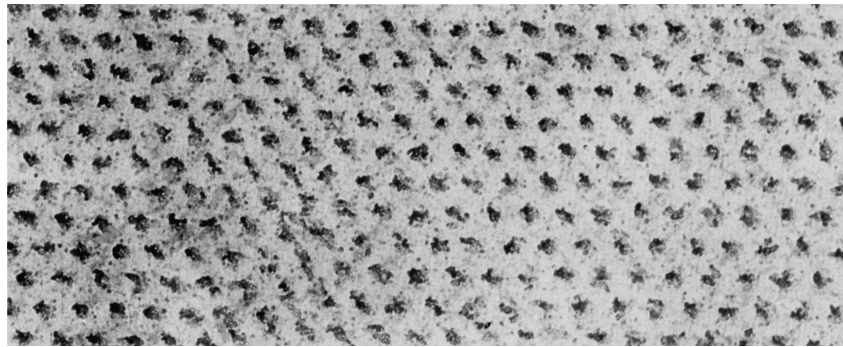


Figure 9: Vortex lattice imaged via electron microscopy using iron-colloid decoration in a Pb + 6.3 at.% In alloy at 1.2 K [Magnification: 8300 $\times$]^[2]

5 Magnetic Susceptibility and Magnetization

In a magnetic field, the magnetisation M of a system is the magnetic moment per unit volume. This is related to the applied magnetic field H linearly by the dimensionless factor magnetic susceptibility, i.e $M = \chi H$ ^[11]¹. The magnetisation and magnetic field relate to each other as:

$$B = \mu_0(H + M) \quad (5.1)$$

For small susceptibilities, this can be written as

$$M = \frac{\chi B}{\mu_0}. \quad (5.2)$$

Here, a **paramagnet** is a substance which has positive susceptibility ($\chi > 0$) and their magnetisation aligns with the applied magnetic field. A **diamagnet** has negative susceptibility ($\chi < 0$) and its magnetisation is in the opposite direction as the applied field. There are no

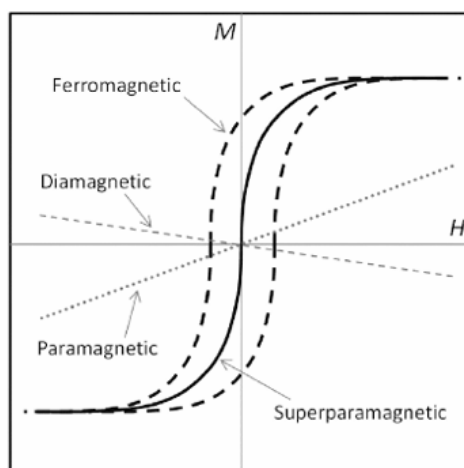


Figure 10: Magnetization curves showing the M vs H relationships for paramagnetic, diamagnetic, and ferromagnetic materials.^[12]

unpaired electrons and the net magnetic moment is zero^[11]. Type I and II superconductors both exhibit perfect diamagnetism since the magnetic fields are expelled from the interior, this results in a magnetic susceptibility of -1. **Ferromagnetic** materials are similar to paramagnetic materials, but its magnetisation M is non-zero even without an applied magnetic field and is much stronger. It can also be called spontaneous magnetism^[11]. The susceptibilities of these different materials are visualised in **Figure 10**.

¹H: Magnetic Field Strength (A/m); B: Magnetic Flux Density (T)

5.1 Critical Field, Temperature, and Current Density

Type-I superconductors have a single critical magnetic field H_c point beyond which beyond which superconductivity is lost entirely. Type-II superconductors are characterised by two critical fields, H_{c1} and H_{c2} . Below H_{c1} , the material exhibits perfect diamagnetism as in the Meissner state. Between the **two critical fields**, magnetic flux penetrates the material in the form of quantised vortices, forming the mixed (Shubnikov) phase, where perfect diamagnetism is no longer maintained^[13]. Above H_{c2} , superconductivity breaks down entirely. In addition to the critical fields, superconducting behaviour is restricted to **temperatures below T_c** , since thermal energy above this point is sufficient to destroy the Cooper pairs responsible for the condensate. A **critical current density J_c** also plays an important role: when the applied current exceeds J_c , the Lorentz force on the Abrikosov vortices overcomes the pinning forces from lattice defects, setting the vortices in motion^[13]. Moving vortices induce an electric field parallel to $\mathbf{J}$, leading to a finite *flux-flow resistance* and energy dissipation even within the mixed state^[13]. Perfect conductivity is therefore only sustained as long as the current remains below J_c .

6 Experimental Methods²

6.1 Four-Point Probe Measurements

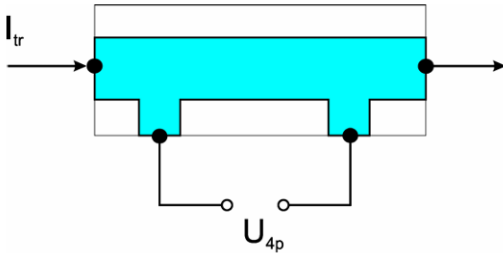


Figure 11: Four point probe measurement setup for measuring the resistance of the given sample.^[3]

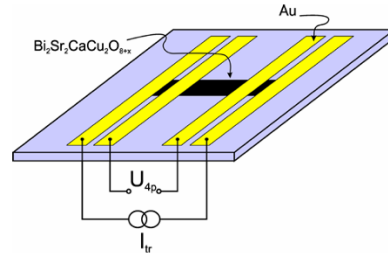


Figure 12: $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+x}$ sample preparation for four-point probe measurements.^[3]

For this experiment, a small sample of the $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+x}$ is removed and fixed on to a sapphire substrate. Four gold contacts are fixed on to the substrate for which thermal evaporation is used here. **Thin aluminium wires** are used for conducting electricity between the gold contacts and the prepared sample. A constant current is passed through the sample and the voltage between the contacts U_{4p} is obtained^[3]. Using a voltmeter of a high internal resistance, one can measure only the small voltage drop between the contacts. Using **Ohm's law**, the resistance can be calculated due to its proportionality with the voltage. If a 2-point method is used, the voltages at the contacts is also measured and results in errors^[3].

The voltage and resistance are then measured at different temperatures by both cooling and heating the sample to produce a hysteresis curve.

6.2 Inductive AC-Susceptibility Measurements

These measurements enables one to find the **critical temperature T_c** and the width of the phase transition ΔT_c . A mutual inductance bridge with three induction coils is used for this

²The whole experimental methods section is based on the experiment manual provided by the university^[3]

measurement^[3]. Two secondary coils are used here which are identical to each other and have equal windings. The first one (reference coil) only experiences the field from the primary coil. Besides this field, the other secondary coil (sample coil) also experiences the field from the superconducting sample. The secondary coils are contained inside the primary coil as shown in Figure 13.

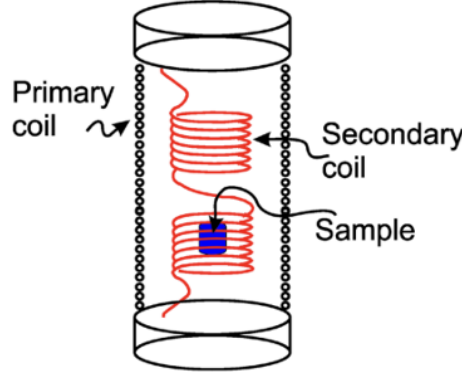


Figure 13: Mutual inductance solenoid setup using primary and secondary coils.^[3]

The application of an alternating current through the primary coil causes an alternating magnetic field inside it. Due to this alternating behaviour, the complex susceptibility and the time dependent magnetic field have to be considered. The external magnetic field follows the form:

$$H_a = H_0 + h_{AC} \cos(\omega t) \quad (6.1)$$

The induced voltage is given by the equation:

$$U = \pi r^2 \mu_0 \left(h_{AC} \omega \sin(\omega t) - \frac{dM}{dt} \right) \quad (6.2)$$

where M is the magnetic momentum, r is the radius of the secondary sample coil, $\sin(\omega t)$ represents the induction signal of the alternating field, and h_{AC} is the amplitude of the alternating field

The magnetic momentum can be written as a Fourier series since it is periodic:

$$M = \langle M \rangle_T + h_{AC} \sum_{n=1}^{\infty} (\chi'_n \cos(n\omega t) + \chi''_n \sin(n\omega t)) \quad (6.3)$$

where χ'_n and χ''_n are the **real and imaginary components** of the magnetic susceptibility. They are written as:

$$\chi'_n = \frac{\omega}{\pi h_{AC}} \int_0^T M \cos(n\omega t) dt \quad (6.4)$$

$$\chi''_n = \frac{\omega}{\pi h_{AC}} \int_0^T M \sin(n\omega t) dt \quad (6.5)$$

Plugging in these terms into the induced voltage equation yields:

$$U = U_0 \left[\sin(\omega t) + \sum_{n=1}^{\infty} n (\chi'_n \sin(n\omega t) - \chi''_n \cos(n\omega t)) \right] \quad (6.6)$$

Using a lock-in amplifier, the $n = 1$ term is isolated from the magnetic momentum. Here, the sin term dominates over the whole induction signal so it has to be removed. This is done so by combining the reference and sample coils in series with opposite windings.

$$U_1^* = U_0 (\chi_1' \sin(\omega t) - \chi_1'' \cos(\omega t)) \quad (6.7)$$

The real and imaginary parts of the susceptibility refer to the **inductive and resistive components**, respectively. The area enclosed by the hysteresis curve of the magnetic flux density is given as:

$$A_H = \oint \mu_0 H_\alpha dB = \int_0^T \mu_0 (H_0 + h_{AC} \cos(\omega t)) \frac{dB}{dt} dt \quad (6.8)$$

The area relates to χ'' as: $A_H = \pi(\mu_0 h_{AC})^2 \chi''$. For normally conducting samples, both components of susceptibility tend to zero, so $U_1^* \approx 0$. For a superconducting sample in the Meissner phase, $\chi' \approx -1$ and $\chi'' \approx 0$, therefore the hysteresis curve is not seen. For type II superconductors, χ' changes while $\chi'' \neq 0$ ^[3].

6.2.1 Deep-Stick Technique

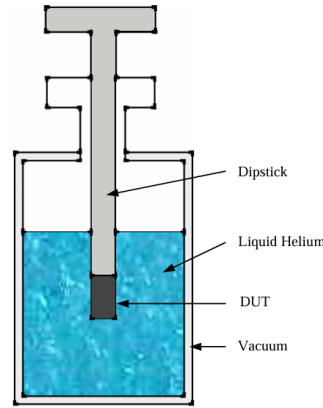


Figure 14: Illustration of the dipstick technique for handling samples at cryogenic temperatures^[14].

The deep stick/dipstick is used when handling samples at low-temperature environments to place them in a cryostat. The sample is placed in a chamber at the end of the dipstick and it can help insert samples in a **liquid helium bath**. Dipsticks can be combined with temperature control devices and diodes for measurement. It allows for the study of the electrical and thermal properties at cryogenic temperatures^[14].

6.2.2 Lock-In Technique

Lock-in amplifiers are useful to extract signals in extremely noisy environments, even extremely weak signals million times lower in amplitude than the noise. The input is multiplied by a **reference signal** which is either provided externally or generated by the lock-in amplifier itself. An **adjustable low-pass** filter is applied to this result, isolating the signal at the required frequency. Frequencies higher than the reference are filtered out^[15].

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 - [16] AI tools such as Claude (Anthropic) and Hawki (FAU) were used in some areas for grammar and language refinement.